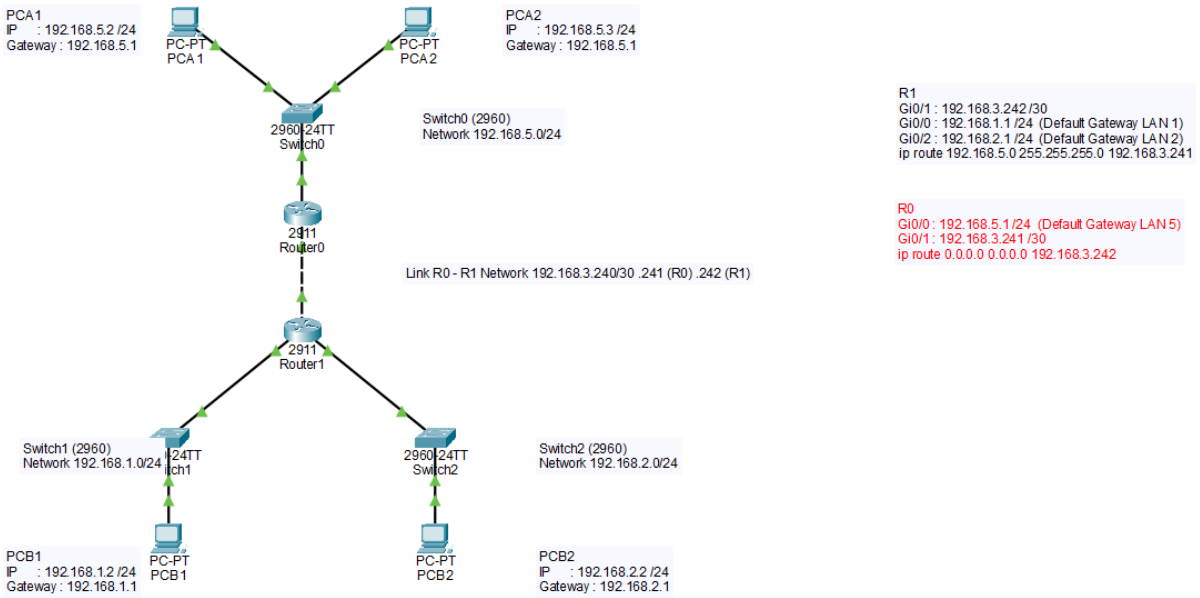
RIP SIMULATION TASK AND DEFAULT GATEWAY

NETWORK CONFIGURATION REPORT

Cisco Packet Tracer

Student Information	Details
Student Name	Saifullah Haidari
University	Politeknik Elektronika Negeri Surabaya (PENS)
Degree Program	Master's Degree in Computer Engineering and Informatics (Robotics)
Course	Computer Networks
Assignment	RIP Simulation Task, Default Gateway
Software	Cisco Packet Tracer
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1. Introduction

This assignment focuses on designing, configuring, and analyzing a small multi-network topology using Cisco Packet Tracer. The topology contains two routers (R0 and R1), three Layer 2 switches, and four end devices. Router R0 connects LAN 5, while Router R1 connects LAN 1 and LAN 2. The routers communicate through a point-to-point link using the 192.168.3.240/30 subnet.

The main purpose is to understand IP addressing, subnet masks, default gateways, and inter-network routing. The supplied configuration uses a static route on R1 and a default route on R0. Connectivity is verified using the ping command and router routing-table inspection.

2. Objectives

- Build the network topology in Cisco Packet Tracer using two routers, three switches, and four PCs.
- Assign IPv4 addresses and subnet masks to all router interfaces and end devices.
- Configure the default gateway on each PC.
- Configure the router interfaces and enable them.
- Configure a static route on R1 and a default route on R0.
- Verify connectivity between devices in different LANs.

3. Network Topology

The network consists of three separate LANs. LAN 5 is connected to Router0 through Switch0. LAN 1 and LAN 2 are connected to Router1 through Switch1 and Switch2, respectively. Router0 and Router1 are connected through a /30 point-to-point network.

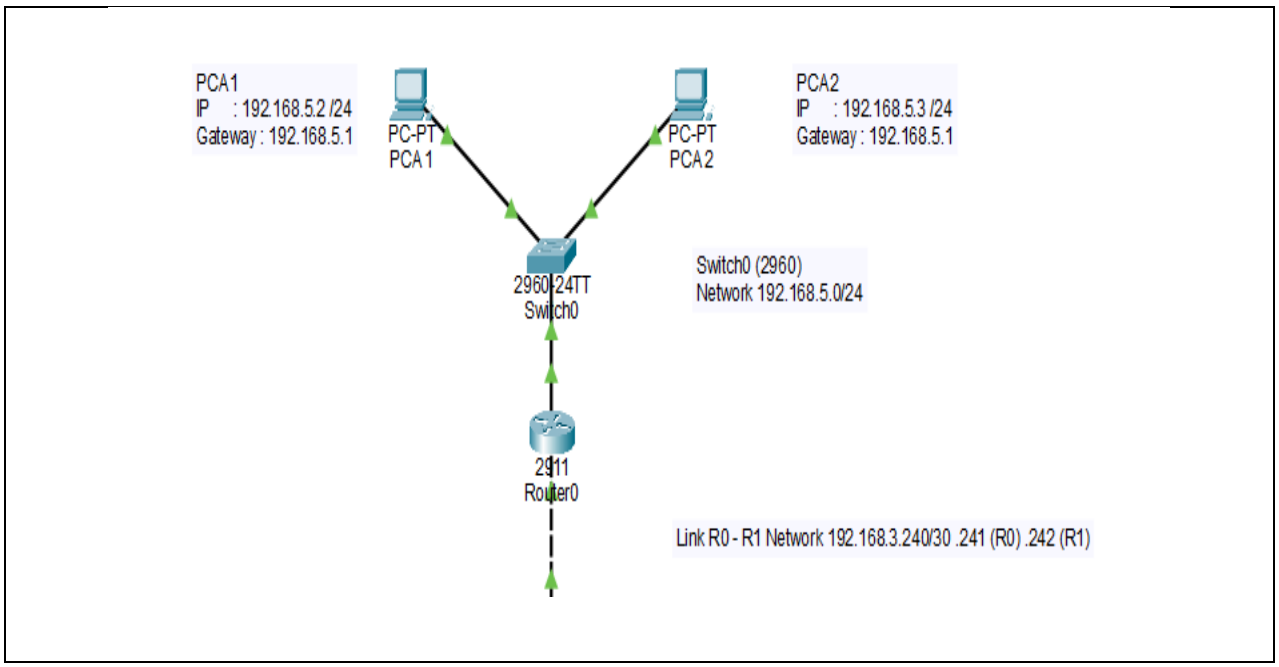


Figure 1. Complete network topology in Cisco Packet Tracer

4. IP Addressing Plan

Network Segment	Network Address	Subnet Mask	Router / Gateway Address
LAN 5	192.168.5.0/24	255.255.255.0	R0: 192.168.5.1

LAN 1	192.168.1.0/24	255.255.255.0	R1: 192.168.1.1
LAN 2	192.168.2.0/24	255.255.255.0	R1: 192.168.2.1
R0–R1 Link	192.168.3.240/30	255.255.255.252	R0: 192.168.3.241; R1: 192.168.3.242

For a /24 network, the subnet mask is 255.255.255.0. For the /30 inter-router link, the subnet mask is 255.255.255.252. In the 192.168.3.240/30 subnet, .240 is the network address, .241 and .242 are usable host addresses, and .243 is the broadcast address.

4.1. PC Address Configuration

Device	IP Address	Subnet Mask	Default Gateway
PCA1	192.168.5.2	255.255.255.0	192.168.5.1
PCA2	192.168.5.3	255.255.255.0	192.168.5.1
PCB1	192.168.1.2	255.255.255.0	192.168.1.1
PCB2	192.168.2.2	255.255.255.0	192.168.2.1

To configure a PC, open the device in Packet Tracer and select Desktop → IP Configuration. Choose Static and enter the IP address, subnet mask, and default gateway according to the table.

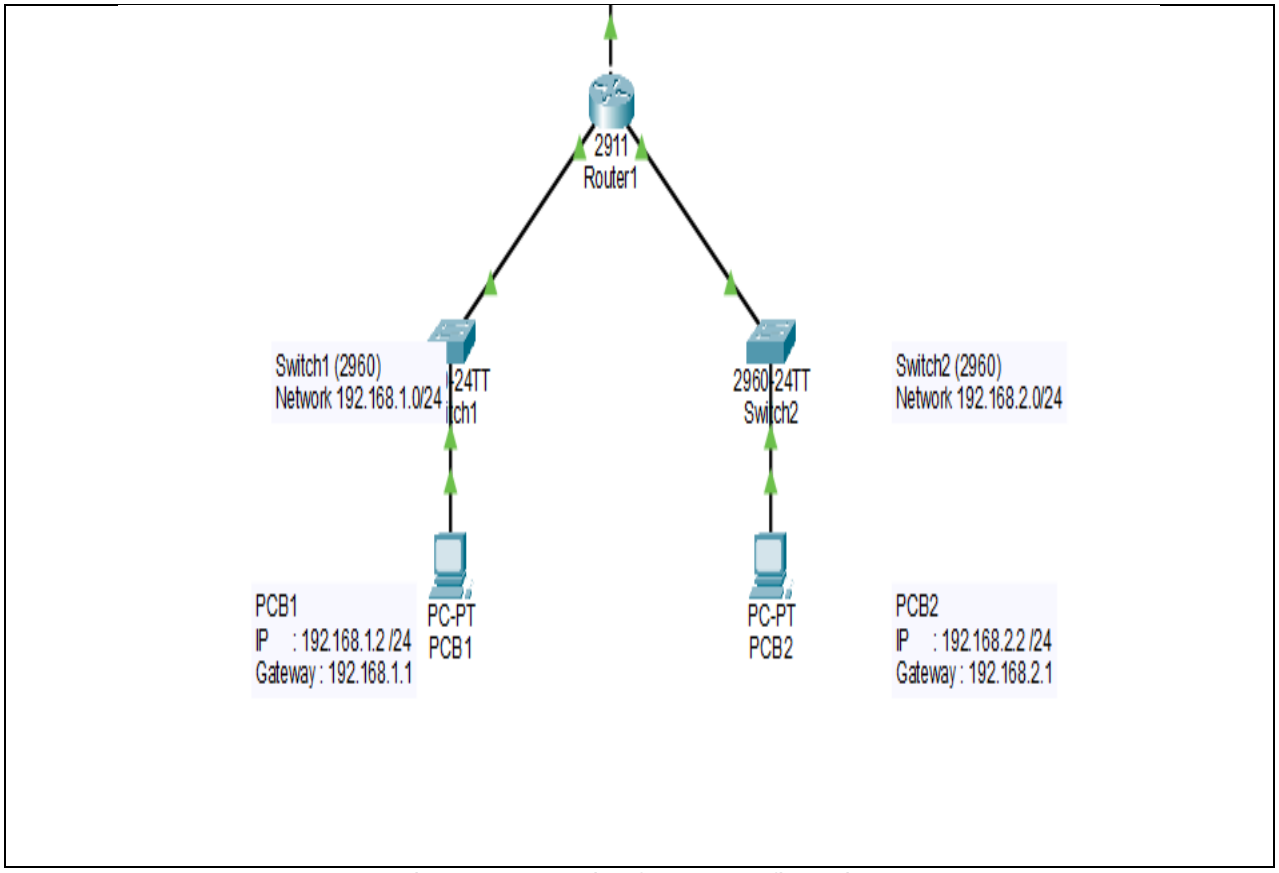


Figure 2. Example of PC IP configuration

5. Router Configuration

5.1. Router0 (R0)

Router0 acts as the default gateway for LAN 5. GigabitEthernet0/0 connects to LAN 5, while GigabitEthernet0/1 provides the point-to-point connection to Router1.

```
enable
configure terminal
hostname R0

interface gigabitEthernet0/0
 ip address 192.168.5.1 255.255.255.0
 no shutdown
 exit

interface gigabitEthernet0/1
 ip address 192.168.3.241 255.255.255.252
 no shutdown
 exit

ip route 0.0.0.0 0.0.0.0 192.168.3.242
end
write memory
```

The default route (0.0.0.0/0) forwards packets for destinations that do not have a more specific matching route to the next-hop address 192.168.3.242, which belongs to Router1.

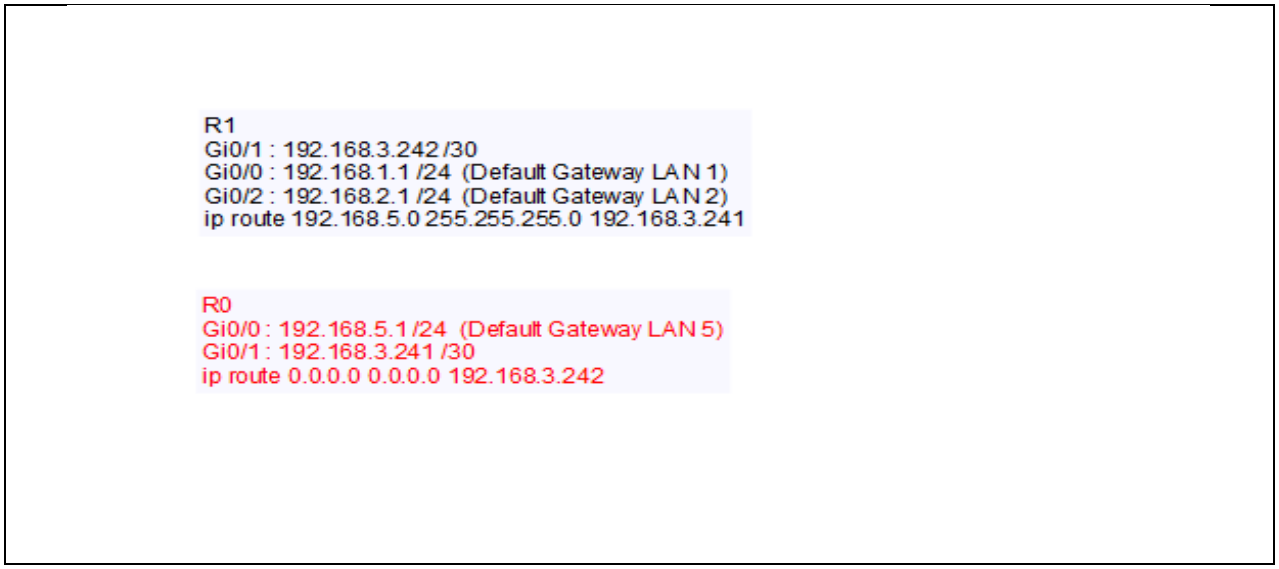


Figure 3. Router0 interface and default-route configuration

5.2. Router1 (R1)

Router1 connects LAN 1 and LAN 2 and provides the default gateway for both networks. GigabitEthernet0/1 connects Router1 to Router0.

```
enable
configure terminal
hostname R1

interface gigabitEthernet0/1
```

```
 ip address 192.168.3.242 255.255.255.252
 no shutdown
 exit

interface gigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
 exit

interface gigabitEthernet0/2
 ip address 192.168.2.1 255.255.255.0
 no shutdown
 exit

ip route 192.168.5.0 255.255.255.0 192.168.3.241
end
write memory
```

The static route tells R1 to forward packets destined for 192.168.5.0/24 to the next-hop address 192.168.3.241, which is the inter-router interface of R0.

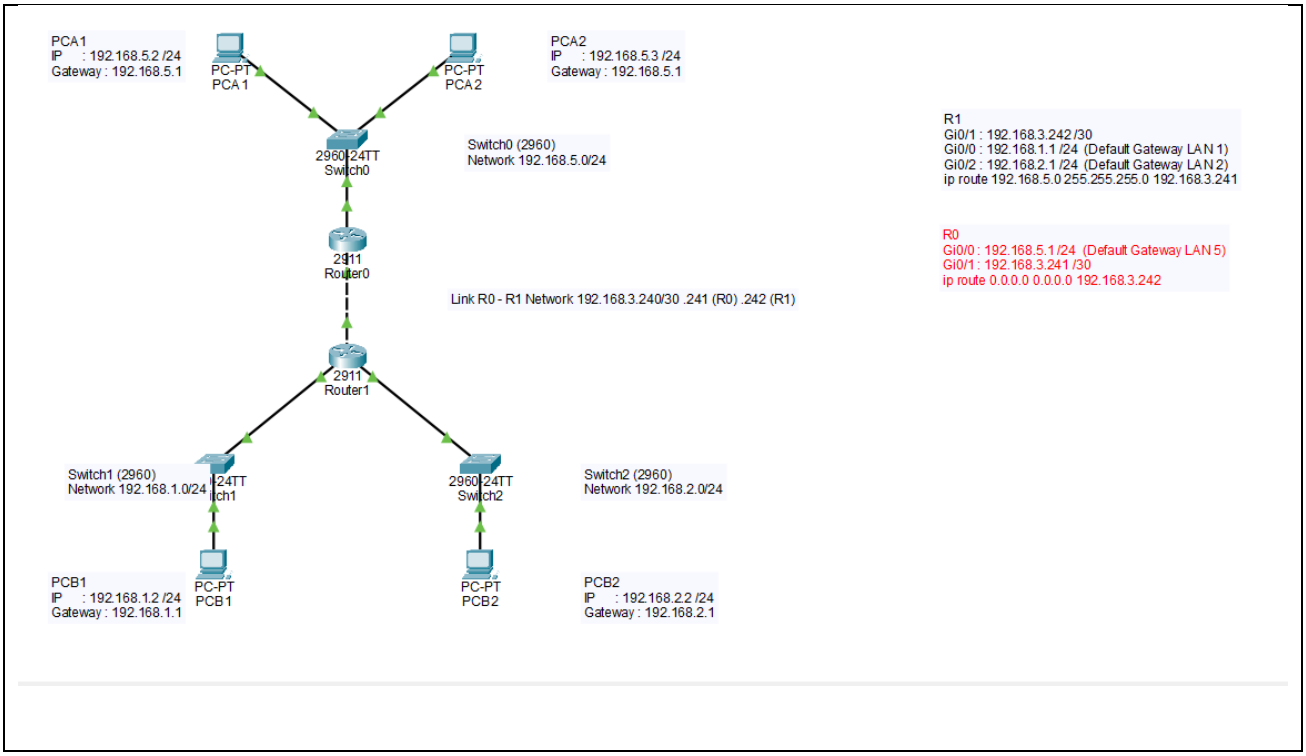


Figure 4. Router1 interface and static-route configuration

6. Default Gateway and Routing Concepts

6.1. Default Gateway

A default gateway is the router interface address that a host uses to send packets to destinations outside its local subnet. For example, PCA1 (192.168.5.2) sends a packet to its default gateway (192.168.5.1) when communicating with PCB1 (192.168.1.2), because PCB1 is not in PCA1's local network.

6.2. Static Route and Default Route

A static route is manually configured to reach a specific destination network. In this topology, R1 uses a static route to reach LAN 5. A default route is used when no more specific route matches the destination. R0 forwards such traffic to R1.

Router	Route Type	Destination	Next-Hop
R1	Static Route	192.168.5.0/24	192.168.3.241
R0	Default Route	0.0.0.0/0	192.168.3.242

Important note: Although the assignment title mentions RIP Simulation, the configuration information provided for this report uses static routing and a default route. It does not enable the RIP protocol. Therefore, this report documents the supplied static/default-routing configuration. If the instructor requires dynamic RIP routing, RIP must also be configured according to the instructor's requirements.

7. Verification and Testing

After configuring the devices, check that the router interfaces are up and that the routing tables contain the expected routes. Use the following commands in the router CLI:

```
show ip interface brief
show ip route
show running-config
```

Use the following ping tests from the PC Command Prompt:

```
PCA1> ping 192.168.5.1
PCA1> ping 192.168.3.242
PCA1> ping 192.168.1.2
PCA1> ping 192.168.2.2

PCB1> ping 192.168.1.1
PCB1> ping 192.168.5.2

PCB2> ping 192.168.2.1
PCB2> ping 192.168.5.3
```

Successful replies indicate that the tested devices can communicate. The actual results must be confirmed by running these commands in Packet Tracer. The screenshots below should show the real output obtained during testing.

8. Network Analysis

Each LAN is assigned a separate subnet, and the routers forward traffic between these subnets. The /24 mask provides a straightforward address range for each LAN. The /30 subnet is suitable for the point-to-point router link because only two usable host addresses are required.

The configured routes provide a path between LAN 5 and the networks connected to R1. R0 forwards traffic without a more specific route to R1 using its default route. R1 uses its static route to send traffic for LAN 5 back to R0. If connectivity fails, check IP addresses, subnet masks, default gateways, interface status, cabling, and the availability of a return route.

9. Conclusion

In this assignment, a network containing three LANs and two routers was designed and configured in Cisco Packet Tracer. IP addresses and default gateways were assigned to the PCs, router interfaces were configured, and static/default routes were added to enable inter-network communication. Connectivity can be verified using ping tests and routing-table inspection. This exercise demonstrates the importance of correct IP addressing, default gateways, and routing configuration in connecting multiple local networks.

END OF REPORT